

also illiquidity risk betas. These are covariances of stock returns with illiquidity measures, like market illiquidity or signed volume.

Estimates of illiquidity risk premiums in the literature range between 1% and 8% depending on which measure of illiquidity is used. However, Ben-Rephael, Kadan, and Wohl (2008) report that these equity illiquidity premiums have diminished considerably—for some illiquidity measures the risk premiums are now zero! In pink sheet stock markets, which are over-the-counter equity markets, Ang, Shtauber, and Tetlock (2013) find an illiquidity risk premium of almost 20% compared to about 1% for comparable listed equities.

Illiquid Assets

There are higher returns to hedge funds that are more illiquid, in the sense that they place more restrictions on the withdrawal of capital (called lockups, see chapter 17) or for hedge funds whose returns fall when liquidity dries up.²⁹ Franzoni, Nowak, and Phalippou (2012) report that there are significant illiquidity premiums in private equity funds—typically 3% (for further details, see chapter 18). In real estate, Liu and Qian (2012) construct illiquidity measures of price impact and search costs for U.S. office buildings. They find a 10% increase in these illiquidity measures leads to a 4% increase in expected returns.

Why Illiquidity Risk Premiums Manifest within but Not across Asset Classes

To my knowledge, we have yet to develop formal equilibrium models explaining the large illiquidity risk premiums within asset classes but not across asset classes.

Perhaps the reason is limited integration across asset classes. There are significant impediments to switching capital and investment strategies seamlessly even across liquid stock and bond markets.³⁰ Investors put asset classes into different silos and rarely treat them consistently as a whole. This happens on both the sell-side, where fixed income, equity desks, and other divisions rarely talk with each other, and on the buy-side, where each asset class is managed by separate divisions. (Canada Pension Plan's factor investing strategy is a notable exception to this, as I discuss in chapter 14.) The potential mispricing of illiquidity across asset classes could reflect institutional constraints, slow-moving capital, and limits to arbitrage.³¹

On the other hand, perhaps asset class illiquidity risk premiums might be small because investors overpay for illiquid asset classes; they chase the illusion of higher returns and bid up the prices of these illiquid assets until the illiquidity premiums to go away. Lack of integrated asset class markets cause investors to make ill-informed decisions for illiquid asset classes. In contrast, within asset classes—especially the more liquid stock and bond markets—illiquidity-shy investors are

²⁹ See Aragon (2007) and Sadka (2010), respectively.

³⁰ See Kapadia and Pu (2012) for evidence of lack of integration across stock and bond markets.

³¹ See Merton (1987), Duffie (2010), and Shleifer and Vishny (1997), respectively.